Basic Number Problems

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1) Count Number of Digits

Problem Statement

Given an integer `n`, return how many digits are present in the number.

Sample Input 1

```
12345
```

Expected Output 1

```
5
```

Sample Input 2

```
9081
```

Expected Output 2

```
4
```

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2) Digit Sum

Problem Statement

Given a number `n`, return the sum of all digits present in the number.

Sample Input 1

```
1234
```

Expected Output 1

```
10
```

Sample Input 2

```
5601
```

Expected Output 2

```
12
```

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3) Reverse a Number

Problem Statement

Given an integer `n`, reverse the digits of the number and print the reversed number.

Sample Input 1

```
12345
```

Expected Output 1

```
54321
```

Sample Input 2

```
980
```

Expected Output 2

```
89
```

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4) Check Prime Number

Problem Statement

Given a number `n`, check whether it is a prime number or not.

Sample Input 1

```
13
```

Expected Output 1

```
Prime
```

Sample Input 2

```
20
```

Expected Output 2

```
Not Prime
```

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5) Return All Prime Numbers from a List

Problem Statement

Given a list of numbers, return all prime numbers present in the list.

Sample Input 1

```
5
2 8 11 15 17
```

Expected Output 1

```
2 11 17
```

Sample Input 2

```
6
4 6 7 10 13 19
```

Expected Output 2

```
7 13 19
```

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6) Print All Prime Numbers up to N

Problem Statement

Given a number `n`, print all prime numbers from `1` to `n`.

Sample Input 1

```
20
```

Expected Output 1

```
2 3 5 7 11 13 17 19
```

Sample Input 2

```
10
```

Expected Output 2

```
2 3 5 7
```

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7) Find All Factors of a Number (Optimized – O(√N))

Problem Statement

Given a number `n`, print all factors of the number using an optimized approach with time complexity close to `O(√N)`.

Sample Input 1

```
36
```

Expected Output 1

```
1 2 3 4 6 9 12 18 36
```

Sample Input 2

```
20
```

Expected Output 2

```
1 2 4 5 10 20
```

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8) Product of Digits

Problem Statement

Given a number `n`, return the multiplication of all digits present in the number.

Sample Input 1

```
345
```

Expected Output 1

```
60
```

Sample Input 2

```
204
```

Expected Output 2

```
0
```